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Drugs quality control
(Theoretical foundation and practical
application): The Coursebook

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CHAPTER OVERVIEW

PART I. Theory

MODULE 1 GENERAL QUESTIONS OF DRUG ANALYSIS AND QUALITY CONTROL

Topic 1 Common approaches for quality control of drugs. Harmonization of pharmacopoeias

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MODULE 1 GENERAL QUESTIONS OF DRUG ANALYSIS AND QUALITY CONTROL

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Topic 1 Common approaches for quality control of drugs. Harmonization of pharmacopoeias

Learning Objectives

- compare and analyze the equal pharmacopoeial monographs (Ph. Eur., USP, JP) and select optimal methods of quality control for different active substances and excipients.

Every medicine is characterized by special requirements of efficiency and safety, which determine its quality. High-grade medicines and their active substances always comply with all regulatory requirements. Medicine quality control is carried out during the whole manufacturing process (at every stage), also when releasing the drug on the market and during its circulation there.

The process of standardization and quality control is implemented in three main areas: identification of medicines, purity assessment (absence of impurities) and assay. Drugs quality indicators together with the methods of their analysis are stated in special regulatory documents (RD). The main one is Pharmacopoeia.

Pharmacopoeia, or **pharmacopoea**, in its modern technical sense, is a book containing directions for the identification of samples and the preparation of compound medicines. It is published by the authority of a government or a medical or pharmaceutical society. Descriptions of preparations are called monographs.

Pharmacopoeia is a collection of official documents (standards and regulations) that sets quality standards for pharmaceutical substances (active pharmaceutical ingredients – API), excipients, diagnostic and medicinal products in different pharmaceutical forms. The provisions of Pharmacopoeia are based on the achievements of pharmaceutical chemistry and pharmaceutical analysis, its criteria, methods and techniques. This document includes instructions for the manufacturing and quality control of pharmaceutical substances, excipients, diagnostic substances and pharmaceutical forms; determines the doses; establishes requirements for medicinal plant materials.

Implementation of Pharmacopoeia standards and requirements combined with the requirements of GMP standards will ensure the quality of pharmaceutical substances and preparations.

Pharmacopoeia has two main sections: general chapters and specific monographs. Specific monographs are devoted to individual substances or finished dosage forms of medicines (USP, JP). Pharmacopoeia monograph is the standard of quality for specific (individual) drug. The general chapters contain information about the methods of analysis, reagents, etc. For example, there are a lot of general chapters in Ph.Eur: Methods of Analysis (Apparatus. Physical and physicochemical methods. Identification. Limit tests. Assays. Biological tests. Biological assays) – Methods in pharmacognosy – Pharmaceutical technical procedures – Materials for Containers –Allergen products – Dosage Forms – Essential oils – Extract – Herbal drugs – Homoeopathic preparations – Immunoserum for human use – Products of fermentation – Radiopharmaceutical preparations – Recombinant DNA technology – Substances for pharmaceutical use – Vaccines for human – Vaccines for veterinary use – Vegetable fatty oils. The State Pharmacopoeia is the official document, which is under state supervision. State Pharmacopoeia is a document of national legislative power, and its requirements are mandatory for all institutions in the state, engaged in the manufacture, storage and use of medicines, including herbal (medicinal plant materials).

In the world pharmaceutical practice, the recognized leaders among the pharmacopoeias are the Pharmacopoeia of Europe, the United States pharmacopoeia and Japan pharmacopoeia.

The Convention developed and the circle of members Pharmacopoeia expanded. Today its membership includes 36 countries and 22 observers. Russia has observer status.

Members of the European Pharmacopoeia Commission are: Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Montenegro, Netherlands, Norway, Poland, Portugal, Romania, Serbia, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, the former Yugoslav Republic of Macedonia, Turkey, United Kingdom and the European Union. The observers to the European Pharmacopoeia Commission are: Albania, Algeria, Australia, Belarus, Brazil, Canada, China, Georgia, Israel, Madagascar, Malaysia, Morocco, Republic of Kazakhstan, Russian Federation, Senegal, Syria, Tunisia, Ukraine, United States of America and WHO (World Health Organization).

Since 1975 if the country wishes to register its products in the European Union it has to use the European Pharmacopoeia. The network of analytical laboratories, controlled by the European Pharmacopoeia Directorate, was organized in 1994. In the same year the attestation procedure for pharmaceutical substances conformity to the demands of European Pharmacopoeia monographs has been introduced.

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Standards of European Pharmacopoeia were widely used in Russian Federation until recently, when they found a reflection in the Russian Pharmacopoeia XIII. European Pharmacopoeia is also being used as a national standard in some countries of the former Soviet Union.

The European Pharmacopoeia is published by the Directorate for the Quality of Medicines & HealthCare of the Council of Europe (EDQM). Commission of European Pharmacopoeia, which includes all members and observers, also regulates work on Ph. Eur. Commission meets three times a year. During the rest of the year, its work is led by elected Bureau. Its functions include the work program adoption, definition of expert groups as well as pharmacopoeia texts approval.

After each monograph has been approved, special group of experts regulates it. The group works according to prescribed control procedures, technology development and revision of pharmacopoeia monographs.

The proposals from national delegations, expert groups, and pharmaceutical industry representatives replenish Ph. Eur. The decision on new monograph inclusion depends on the therapeutic effect of drug, widespread of its use, on the number of countries that have ratified it and on its proven quality.

The proposals from national delegations, expert groups, and pharmaceutical industry representatives replenish Ph. Eur. The decision on new monograph inclusion depends on the therapeutic effect of drug, widespread of its use, on the number of countries that have ratified it and on its proven quality.

US Pharmacopoeia Convention is one of the most influential pharmacopoeia organizations in the world. Its quality standards are not only introduced in the US and Canada, but are also put into service in a number of countries actively producing drugs (India, China, etc). USA Pharmacopoeia Convention is a nonprofit organization, but its financial resources and influence are very high over the world. Convention publishes a number of periodicals devoted to drug quality standards, develops and sells worldwide a rich collection of pharmacopoeia chemical reference materials, which quality is recognized all over the world.

Japanese Pharmacopoeia (JP) is another leading international pharmacopoeia. The new edition is published every five years. Current edition (JP 17) came in 2016. JP is published not only in Japanese but also in English.

British Pharmacopoeia (BP) has been published since 1864, the current edition – BP 2019. BP operates in the United Kingdom simultaneously with the European Pharmacopoeia. BP monographs for substances are practically the same as the corresponding Ph. Eur. monographs. BP also contains monographs for finished dosage forms of medicines.

In addition, the World Health Organization since 1950 publishes the International Pharmacopoeia. Today it is represented by the 8th edition, published in 2018. The main objective of the International Pharmacopoeia is to maintain the medicines quality in developing countries. In the absence of state resources to develop their own pharmacopoeia, these countries can take the International Pharmacopoeia as a national one.

The XIV edition of Russian Federation (Ph.RF) State Pharmacopoeia was published in 2018. Monographs describe physical and chemical characteristics of different dosage forms, analysis methods, demands for drug quality indicators. Monographs defines the quality requirements of medicines, active pharmaceutical substances and excipients.

Harmonization of pharmacopoeias. Increased facilities for travel have brought into greater prominence the importance of an approach to uniformity in the formulae of the more powerful remedies, in order to avoid danger to patients when a prescription is dispensed in a different country from that in which it was written. The first attempts were made by international pharmaceutical and medical conferences to settle a basis on which an international pharmacopoeia could be prepared, but due to national jealousies and the attempt to include too many preparations nothing has yet been achieved.

Globalization processes indicated the need for harmonization of medicines quality requirements. Harmonisation of various countries pharmacopoeia requirements is determined by ICH (www.ich.org) – the International Conference on Harmonization of

Medicines Quality and the International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use. The World Health Organization and three leading Pharmacopeias (Ph. Eur, USP, JP) are directly involved in ICH activity. The purpose of ICH is to develop common requirements for standardization, quality control, efficiency, safety, production and registration of medicines. Representatives from three leading Pharmacopeias meet twice a year since 1990 in the Pharmacopoeial Discussion Group to try to work towards «compendial harmonisation». Specific monographs are proposed, and if accepted, proceed through the stages of review and consultation. Adoption of a common monograph provides a common set of tests and specifications for a specific material. Not surprisingly, this is a slow process.

Though formerly printed there has been a transition to a situation where pharmaceutical information is available as printed volumes and on the internet. The rapid increase in knowledge renders necessary frequent new editions, to furnish definite formulae for preparations that have already come into extensive use in medical practice, so as to ensure uniformity of strength, and to give the characters and tests by which their purity and potency may be determined. However each new edition requires several years to carry out numerous experiments for devising suitable formulae, so that current pharmacopoeia is never quite up to date.

General monographs of Ph. Eur. «Pharmacopoeial harmonization» states: «It provides information on the degree of harmonization and consequently interchangeability of various general chapters and monographs of the European Pharmacopoeia and those of the Japanese Pharmacopoeia and United States Pharmacopoeia».

The European Pharmacopoeia Commission recognizes the utility of working with other pharmacopoeial bodies to develop harmonized monographs and general chapters. Such harmonization is fully compatible with the declared aims of the Commission and has benefits of different kinds, notably the simplification and rationalization of quality control methods and license in procedures. Such harmonization also enhances the benefits of the work of the International Conference on Harmonization (ICH) and the Veterinary International Cooperation on Harmonization (VICH) since some of the guidelines developed depend on pharmacopoeial general chapters for their application.

Harmonization is carried out in the Pharmacopoeial Discussion Group (PDG), in which the European Pharmacopoeia, the Japanese Pharmacopoeia and the United States Pharmacopoeia are associated. Where harmonization of monographs is carried out, the aim is to arrive at identical requirements for all attributes of a product. For some products, it can be extremely difficult to achieve complete harmonization, for example because of differences in legal status and interpretation.

Anniversary of the Pharmacopoeia Convention and marks the occupation of the new purpose – designed headquarters building of the EDQM, the European Directorate for the Quality of Medicines & Health Care, in Strasbourg.

The monographs of the Pharmacopoeia, both specific and general, together with other texts made mandatory by virtue of reference in monographs, are applicable throughout the 37 Member States including the European Union itself, which is also a signatory to the European Pharmacopoeia Convention. This means that the European Pharmacopoeia holds a special place in the regulatory processes within the European Union, its text being made mandatory or given ‘mandatory’ applicability by virtue of reference in European Council Directives. In addition to the 37 signatories to the European Pharmacopoeia Convention, there are also a large number (20) of observer countries. Consequently, the quality standards developed through the Pharmacopoeia have an impact on the quality of medicinal products and substances used across a large part of the globe.

The cooperation between the experts from industry, academia, regulatory authorities and official government laboratory scientists represents the pinnacle of scientific cooperation to produce a high standard of technical monographs and chapters.

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Topic 2 Pharmaceutical analysis data treatment. Validation of analytical procedures

Learning Objectives

to get an idea about the rules of pharmaceutical analysis data treatment and reporting, the scope and process of validation of analytical procedures, consolidate knowledge in problem solving and laboratory work.

Analytical signals treatment and transformation into the analytical data concerning the nature and amount of a substance, its chemical structure or spatial location of the sample are important components of the pharmaceutical analysis. The result of analysis not statistically treated is of little value.

According to the pharmacopoeial and GMP requirements analytical procedures used for drugs quality control should be validated.

The purpose of validation process is to confirmation experimentally that the analytical procedure provides relevant and reliable information about the object of analysis, and is suitable for its intended purpose.

All quantitative tests (assays) should be validated, including impurities limit tests. Identification tests are validated if it is necessary to confirm their specificity.

Validation of analytical procedures (especially those that will be used as the official process) is regulated by series of standards issued by National and International organizations:

- Russian State Pharmacopoeia XII OFS 42-0113-09 «Validation of Analytical Procedures»
- USP <1225> Validation of Compendial Procedures
- ICH (Q2A, Q2B) Validation of Analytical Procedures: Text and Methodology.

Validation of analytical methods should be evaluated on the following characteristics – the validation parameters:

- Accuracy;
- Precision;
- Specificity;
- Detection Limit;
- Quantitation Limit;
- Linearity;
- Range;
- Robustness.

Guidelines for choosing the parameters of validation are shown in [Table 2.1](#).

[Table 2.1](#)

Revalidation of an analytical procedure is performed in case of:

- changes in drug manufacturing process (the object of analysis);
- changes in the drug's composition;
- changes in the previously approved analytical procedure.

Trueness with regard to a set of measurements involves a combination of two components: **Accuracy** and **Precision**.

Accuracy is characterized by the closeness of the test result obtained by the procedure to the true value ([Fig. 2.1](#)). The value of the *bias* (systematic error) is an indicator of the accuracy. An analytical procedure to be validated is recognized accurate, if the values accepted as conventionally true are within the confidence intervals of the relevant average tests results obtained by this procedure.

[Fig. 2.1]

There are different approaches for assessing the accuracy of quantitative tests (assays):

1. analysis by validated procedure using Reference Standards (RS) or model mixtures to which known quantities (concentration) of the drug substance to be analyzed;
2. comparison of the results of the proposed analytical procedure with those of a second well-characterized procedure, the accuracy of which is stated and/or defined;

3. consideration the results linearity of proposed analytical procedure: if the absolute term of equation $y = b \times x + a$ is not statistically different from zero, then the use of this procedures give results free of bias.

Precision is characterized by the scattering of the results obtained by the procedure to the average value (Fig. 1). This characteristic depends only on random factors and not connected with the true value to be measured. The Measure of this scattering is the *standard deviation* of individual measurement obtained for a sufficiently large amount of sampling.

Precision is assessed for all quantitative tests (assay) using the results of at least three determinations for 3 levels of values to be measured (lower, middle and upper), which are within the range. In many cases, the precision can be evaluated based on the experimental data by least square method.

Precision study is performed using homogeneous samples and can be assessed in three ways:

- as repeatability;
- as within-laboratory (intermediate) precision;
- as between-laboratory reproducibility (precision).

The results of analytical procedures estimation for each of the options are usually characterized by the corresponding value of the Standard Deviation of individual determination with the number of degrees of freedom.

Usually the repeatability of the results obtained by the original procedure is estimated in the development process. In case of inclusion of the developed analytical procedure in normative document, within-laboratory (intermediate) precision should be estimated additionally. Between-laboratory reproducibility (precision) of analytical procedure should be estimated with its proposed inclusion in the General Pharmacopoeial Article, Pharmacopoeial Article or RS normative document.

Repeatability is assessed by individual measurements results obtained in the same laboratory conditions (the same operators, the same equipment or reagents within a short period of time) (Fig. 1). Repeatability is estimated by the relative standard deviation (RSD), which should not exceed 2%.

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CHAPTER OVERVIEW

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Laboratory Practice Session 01

COMMON APPROACHES FOR QUALITY CONTROL OF DRUGS. HARMONIZATION OF PHARMACOPOEIAS

I. Questions and tasks for discussion:

1. Explain the differences between the terms – «drug», «medicine», «pharmaceutical substance», «dosage form».
2. What are the regulations of medicines quality and at what stages of drug preparation it is necessary to use them?
3. What is Pharmacopeia, State Pharmacopeia?
4. Explain differences between the two types of regulations – «individual monographs» and «general monographs».
5. What do you know about the world's leading pharmacopoeias (Ph. Eur., USP and JP)?
6. List the features of the International Pharmacopoeia.
7. List the main features of the British Pharmacopoeia.
8. What is the harmonization of the pharmacopoeias?
9. Clarify the abbreviations: ICH, EDQM, PDG, PMDA, USP? What are the main purposes of these organizations?

II. Examine and compare general articles of leading world pharmacopoeias.

Examine the proposed articles (appendix 1) from different pharmacopoeias: European, Japanese and U.S. Pharmacopoeia. Pay attention to the article title and to the pharmacopoeial section, which includes this article. Find similarities and differences in the methods of analysis. Fill the table according to the plan:

1. Identify the purpose of the test – which drug qualities does it determine?
2. Depict the general scheme of the test, write (where possible) chemical reaction equation.
3. Indicate test conditions: reagent concentration, temperature, time, etc.
4. Make a conclusion about the harmonization degree of these pharmacopoeial articles.

No	Ph. Eur. (7 th ed.)	JP (16 th ed.)	USP (31 th ed.)
1	2. Methods of analysis 2.4. Limit tests 2.4.13. Sulfates	1. Chemical Methods 1.14. Sulfate limit test	Chemical tests and assays, Limit tests 221. Chloride and Sulfate
2	-	1. Chemical Methods 1.15 Readily carbonizable substances test	Chemical tests and assays, Limit tests 271. Readily carbonizable substances test
3	2. Methods of analysis 2.4. Limit tests 2.4.14. Sulfated ash	2. Physical Methods 2.44. Residue on Ignition test	Chemical tests and assays, Limit tests 281. Residue on Ignition test
4	2. Methods of analysis, 2.2. Physical and physicochemical methods 2.2.32. Loss on drying	2. Physical Methods 2.41. Loss on drying test	Physical tests and determinations 731. Loss on drying

№	Ph. Eur. (7 th ed.)	JP (16 th ed.)	USP (31 th ed.)
5	Loss on ignition – test is included in monographs	2. Physical Methods 2.43. Loss on ignition test	Loss on ignition – test is included in monographs

III. Study the table below. Find and represent in writing form similarities and differences in Pharmacopoeias

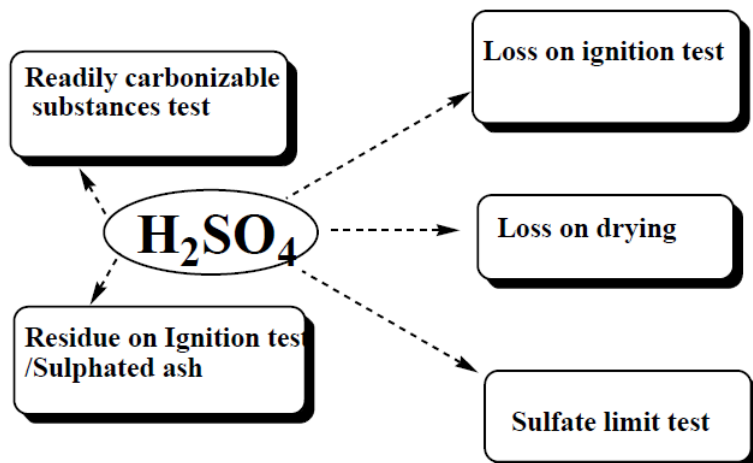
European Pharmacopoeia (Ph. Eur.) <i>Inter-governmental</i>	United States Pharmacopoeia <i>Independent of government</i>	Japanese Pharmacopoeia <i>Governmental</i>
The European Pharmacopoeia of the Council of Europe is a pharmacopoeia, listing a wide range of active substances and excipients used to prepare pharmaceutical products in Europe. It includes more than 2000 specific and general monographs, including various chemical substances, antibiotics, biological substances; Vaccines for human or veterinary use; Immunosera; Radiopharmaceutical preparations; Herbal drugs; Homoeopathic preparations and homoeopathic stocks. It also contains Dosage forms, General monographs, Materials and Containers, Sutures; 268 General methods with figures or chromatograms and 2210 reagents are described. The monographs give quality standards for all the main medicines used in Europe. All medicines sold in the 36 Member States of the European Pharmacopoeia must comply with these quality standards so that consumers have a guarantee for products obtained from pharmacies and other legal suppliers. The European Pharmacopoeia is developed by the European Directorate for the Quality of Medicines (EDQM) and is a part of the Council of Europe, Strasbourg, France. It has been created by the <i>Convention on the elaboration of a European Pharmacopoeia</i> from 1964. Currently there are 37 members of the European Pharmacopoeia Commission: Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland,	It is the official pharmacopoeia of the United States, published dually with the National Formulary as the USP-NF. The United States Pharmacopeial Convention (usually also called the USP) is the nonprofit organization that owns the trademark and copyright to the USP-NF and publishes it every year. Prescription and over-the-counter medicines and other health care products sold in the United States are required to follow the standards in the USP-NF. USP also sets standards for food ingredients and dietary supplements. The United States Pharmacopoeia and The National Formulary (USP-NF) is a book of public pharmacopeial standards. It contains standards for (chemical and biological drug substances, dosage forms, and compounded preparations), excipients, medical devices, and dietary supplements. USP-NF is a combination of two compendia, the United States Pharmacopoeia (USP) and the National Formulary (NF). Monographs for drug substances, dosage forms, and compounded preparations are featured in the USP. Monographs for dietary supplements and ingredients appear in a separate section of the USP. Excipient monographs are in the NF. A monograph includes the name of the ingredient or preparation; the definition; packaging, storage, and labeling requirements; and the specification. The specification consists of a series of tests, procedures for the tests, and acceptance criteria. These tests and procedures require the use of official USP Reference	Japanese Pharmacopoeia (JP) is a book of official pharmaceutical standards in Japan, prepared by the JP secretariat in PMDA and established by the MHLW. It is evaluated and reviewed in expert committees periodically held by the JP secretariat. Pharmaceuticals and Medical Devices Agency (PMDA) is an incorporated administrative agency, established under the Act on PMDA in 2004. It conducts scientific and technical evaluations and reviews on pharmaceuticals and medical devices under the entrustment and medical devices, under the entrustment from the Ministry of Health, Labor and Welfare (MHLW). 17th edition

European Pharmacopoeia (<i>Ph. Eur.</i>) <i>Inter-governmental</i>	United States Pharmacopoeia <i>Independent of government</i>	Japanese Pharmacopoeia <i>Governmental</i>
Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Montenegro, Netherlands, Norway, Poland, Portugal, Romania, Serbia, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Republic of Macedonia, Turkey, United Kingdom, and the European Union. In these countries and the European Union the Ph. Eur. is the official pharmacopoeia. Additional local pharmacopoeias may exist (e. g. in United Kingdom and Germany). Not all of these countries are member states of the European Union. Observers from 20 member and non-member states of the Council of Europe and international organizations take part on sessions of the European Pharmacopoeial Commission: Albania, Algeria, Australia, Belarus, Brazil, Canada, China, Georgia, Israel, Madagascar, Malaysia, Morocco, Kazakhstan, Russian Federation, Senegal, Syria, Tunisia, Ukraine, United States of America, and the WHO. 7th edition	Standards. Medicinal ingredients and products will have the stipulated strength, quality, and purity if they conform to the requirements of the monograph and relevant general chapters. Tests and procedures referred to in multiple monographs are described in detail in the USP–NF general chapters. The General Notices provide definitions for terms used in the monographs, as well as information that is necessary to interpret the monograph requirements. USP is proposing to revise the General Notices for the USP and NF. The U.S. Federal Food, Drug, and Cosmetics Act designates the USP–NF as official compendia for drugs marketed in the United States. A drug product in the U.S. market must conform to the standards in USP–NF to avoid possible charges of adulteration and misbranding. 31th edition	

IV. Read the sentences. Say, if it is true (T) or false (F)?

1. <i>Readily Carbonizable Substances Test</i> is a method to examine the minute impurities contained in drugs, which are readily colored by addition of concentrated acetic acid.	
2. <i>Loss on ignition test</i> is carried out for drugs of organic nature, which lose a part of the components or impurities during ignition.	
3. <i>Loss on Ignition Test</i> is a method to measure the loss in mass when the sample is ignited under the conditions specified in monograph.	
4. To conduct <i>Loss on drying test</i> you can dry the substance in exsiccator at 101,3 kPa or under vacuum (different pressure, sometimes with heating) and in the oven at temperature, specified in monograph.	
5. In <i>Sulfate limit test</i> the solution of BaCl ₂ is used as a reagent, and the standard solution of H ₂ SO ₄ is used as a control solution.	
6. <i>Residue on ignition test (Sulfated ash)</i> is necessary stage of heavy metals limit test. <i>Residue on ignition test</i> is carried out at T > 400°C in the presence of H ₂ SO ₄	

V. Look at the scheme. Cross out the direction where sulfuric acid is not required during the test:



VI. Self-test:

1. What is pharmacopoeias harmonization (negotiation)?
 - a. The development and implementation of standards of medicines as the basis of their safety and effectiveness.
 - b. The free movement of medicines in Europe.
 - c. Quality assurance of drug substances and excipients when importing or exporting from Europe.
 - d. Coordinated control of production and quality of medical products in connection with the free movement of medicines.
2. Pharmacopoeia harmonization – is harmonization of requirements for quality of medicines between:
 - a. European countries.
 - b. Europe and Russia.
 - c. Britain, France and the USA.
 - d. European Pharmacopoeia, the U.S., Japan and other countries pharmacopoeias.
3. According Ph.Eur. in the test «loss on drying» the substance is dried under conditions:
 - a. In an exsiccator over a reagent to absorb water.
 - b. In vacuum.
 - c. In vacuum at a given temperature.
 - d. In an oven at a certain temperature.

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Laboratory practice session 02

PHARMACEUTICAL ANALYSIS DATA TREATMENT. VALIDATION OF ANALYTICAL PROCEDURES

I. Questions and tasks for discussion:

1. The terms «error in analysis» and «uncertainty of measurement». Classification and control of errors.
2. Statistical treatment and reporting of results. Confidence interval.
3. What are the two main validation parameters? What types of errors do they characterize?
4. How many measurements must be performed to obtain statistically reliable data during the validation process?
5. What are the main ways for achieving the accuracy?
6. Give your definition of «repeatability» and «reproducibility».

II. Project «Validation of Analytical Procedure».

Description

Pharmaceutical company is planning to sell generic. To confirm that the manufactured product meets quality requirements, the company orders a testing laboratory to develop an analytical procedure of quantitative determination of acetaminophen (assay).

For the research laboratory the possibility to get this order depends on the reliability and on validity of its analytical procedures.

So, if the first task for you is to select and evaluate regulation validation parameters in terms of ICH, the second task will be to assess the ability of testing laboratory to provide competent performance validation.

When the Lab receives from the Customer a list with approved ICH validation parameters required for the HPLC determination of acetaminophen, Lab should make sequence (order) of their evaluation. Along the way, set the appropriate limits for specification parameters (Table 1).

Table 1 Necessary validation parameters & specifications

Validation parameter	Acceptance criteria (norms)
Specificity	Homogeneity of the peaks, resolution coefficient (R_s) $>1,5\%$
Linearity	Correlation coefficient (r) $> 0,998$
Range	Homogeneity of dispersion confirmed by Fischer method
Accuracy	Recovery is 98–102% (spiked placebo, standard addition of analyte to placebo)
Repeatability	RSD $< 1,5\%$

Assessment of relevant validation parameters is performed according to the scheme shown in Figure 1. Selected validation parameters will be determined after the necessary research is conducted.

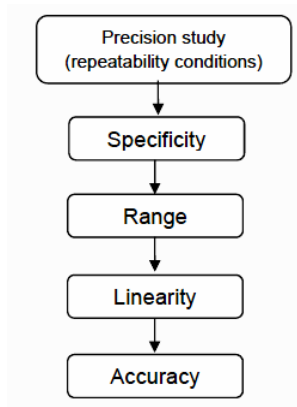


Figure 1: Sequence of validation parameters determination

1. Precision study (repeatability conditions).

HPLC separation and determination of acetaminophen in testing solutions was performed six times in repeatability conditions. The mass concentration of acetaminophen in testing solutions equals 30 mg/l. Peak areas are shown in the table 2.

Table 2 HPLS assay results: chromatographic peak areas

Run	Peak area, u
1	63 120
2	62 601
3	63 887
4	62 714
5	62 986
6	63 407

Task

Using relevant software or calculator estimate the Standard Deviation and RSD, % (coefficient of variation).

2. Specificity Study.

The expected value of mass concentration of acetaminophen in the sample was 30 mg/l. The Figure 2 shows the HPLC chromatogram obtained with the assay of sample. It can be seen that except the main peak of acetaminophen, there is a peak of relative component. Homogeneity of the peaks was performed by the means of chromatography-mass spectrometry to assess the specificity. The results showed that the peaks belong to pure substances, the homogeneity of the peaks in this case was confirmed.

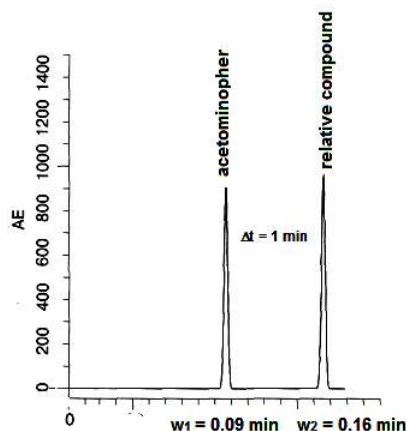


Figure 2: Chromatogram of standard sample with known acetaminophen concentration

Task

Calculate resolution (R_s), which defines the degree of separation. Use chromatographic parameters (Fig. 2.8) and equation:
[equation]

where Δt – distance between the peaks;

w – width of each peak at the base as shown in Fig. 2.8.

3. Range Study.

Six calibration solutions with acetaminophen concentration from 20 to 45 mg/l were prepared to test the range for homogeneity dispersion. The measured peak areas are shown in Table 3.

Table 3 HPLS results for calibration solutions

Peak area, u (Cacetaminophen = 20 mg/l)	Peak area, u (Cacetaminophen = 45 mg/l)
45 530	99 472
45 209	98 912
44 889	99 130
45 022	99 899
45 410	99 451
44 706	98 971

Homogeneity of dispersion at the end of the calibration curve was performed by Fisher method.

Task

Calculate Fisher statistics (ζ) using equation:

[equation]

where S_1 , S_2 – dispersions of the two series of chromatographic data, $S_1^2 > S_2^2$.

If the confidence level $P=95\%$ and the number of degrees

[equation]

critical value equals 5,05. Make a conclusion about the homogeneity of dispersions of the two series of chromatographic data.

4. Linearity Study.

To estimate the concentration (mass) of acetaminophen in the test solution calibration solutions were prepared and the area of the peaks was measured using relevant software. The HPLS results are presented in Table 4.

Table 4 HPLC analysis of the calibration solutions of acetaminophen

$C_{\text{acetaminophen}}$, mg/l	Peak area, u
20	45 450
25	55 290
30	65 165
35	76 254
40	88 019
45	99 309

Task

Using data shown in Table 4 and Microsoft Excel construct a calibration linear curve « $C_{\text{acetaminophen}}$ – peak area» and obtain linear regression equation. Calculate correlation coefficient (r).

5. Accuracy Study.

To determine the recovery, % three placebo samples were spiked by 24, 30 and 36 mg of paracetamol respectively («3 × 3 method»). Concentrations of paracetamol obtained by HPLC analysis are shown in the Table 5.

Table 5 HPLC analysis of the calibration solutions of acetaminophen

Run	Added amount of acetaminophen, mg	Obtained amount of acetaminophen, mg			Mean, mg
1	24	20,2	21,6	21,5	21,1
2	30	26,8	27,5	26,6	27,0
3	36	32,5	32,6	31,6	32,2

Task

Using data shown in Table 6 calculate recovery, % by the equation:

[equation]

6. Results and reporting.

Summarize the results of validation studies in the table 6 Check the acceptance criteria: are the established norms fulfilled? Fill in the last column of the table (Yes/No).

Answer the question: Would you trust this laboratory to perform an assay?

Table 6 Validation results

Validation parameter	Acceptance criteria	Result	Acceptance criteria met (Yes/No)
Linearity	correlation coefficient $r > 0,998$	correlation coefficient $r =$	
Range	Fischer method ($P=95\%$, $n=6$) statistics over the table value	Statistics value = Critical value =	
Specificity	homogeneity of the peaks is observed when resolution $R_s > 1,5\%$	homogeneity of the peaks is observed when $R_s =$	
Accuracy	Recovery = 98–102% (spiked placebo method, standard addition of analyte to placebo)	Recovery =	

Validation parameter	Acceptance criteria	Result	Acceptance criteria met (Yes/No)
Repeatability	RSD < 1,5%	RSD =	

III. Self-test:

1. Metrological certification of analytical procedure is – ...
 - a. its qualification.
 - b. its validation.
 - c. laboratory verification of the its suitability.
 - d. within-laboratory quality control.
2. Ways to identify and eliminate systematic errors in analysis ...
 - a. varying the sample size.
 - b. use standard additions of analytes.
 - c. use of reference materials.
 - d. application of all mentioned ways.
3. Validation of analytical procedure is performed by evaluation the following characteristics (parameters):
 - a. trueness
 - b. repeatability
 - c. robustness
 - d. selectivity (specificity)
 - e. Linearity
 - f. detection limit
 - g. quantitation limit
4. Random errors ...
 - a. characterize the precision of analysis.
 - b. can be estimated by statistical treatment methods.
 - c. can not be estimated by statistical treatment methods.
 - d. have causes that can not be established definitely.
5. Systematic errors ...
 - a. characterize the accuracy of analysis.
 - b. characterize statistically significant difference between the average and the true value.
 - c. can be identified using reference standards, certified metrological procedures or by standard additions method.
6. Precision evaluation includes assessing:
 - a. trueness
 - b. repeatability
 - c. accuracy
 - d. limit of detection
 - e. reproducibility

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Laboratory practice session 03

THERMAL ANALYSIS IN LEADING PHARMACOPOEIAS. MELTING POINT DETERMINATION OF API

I. Questions and tasks for discussion:

1. What is Thermal Analysis? What types of thermal analysis are described in the pharmacopoeias: Ph. Eur., JP, USP?
2. What quality levels of drug products can be examined by means of these methods?
3. What is the difference between the common and differential type of thermal analysis? Describe the main advantages and disadvantages.
4. Figure the general view of the thermoanalytical curve.
5. Characterize DSC method. How does DSC differ from the DTA?
6. Comment the use of TGA for determination «loss on drying», «loss on ignition», «water content», «residue on ignition ». Figure thermoanalytical curves corresponding to each of them.

II. Analytical work with pharmacopoeial general chapters «Melting point» (as an indicator of the quality of API).

Examine the proposed Eur. Ph., JP, Int. Pharmacopoeial general chapters – «Melting point determination». Complete the table.

JP	Ph. Eur.	Int. Ph.
1. Capillary method is applied for easily dispersible crystalline substances...	1. Capillary method	1. Capillary method
2. Open capillary method (both ends of the capillary are open) is applied to substances such as fats, fatty acids...	2. Open capillary method	2. Open capillary method
3. Drop point	3. Instantaneous method	-
-	4. Drop point	-

III. Lab practice:

Determine the melting point of one of the proposed APIs. According to the results of the test fill in the form:

Melting point determination of API by capillary method:
 Normative document (ND)
 2.2.14. MELTING POINT – CAPILLARY METHOD _____
 Testing sample _____
 Batch/Lot _____
 Manufacturer _____
 Limits _____

Apparatus and materials:
 1. Instrument _____
 2. Capillaries _____
 3. Sample bottle _____
 4. Desiccator with desiccant _____
 5. Glass tubes _____
 6. Mortar and pestle _____

Procedure description:
 Find in Eur. Ph.
 2.2.14. MELTING POINT – CAPILLARY METHOD

Test results

N ^o capillary with sample	T _{meas.} , °C	[T bar], °C	$\Delta = T_{\text{meas}} - T_{\text{cert.}}, ^\circ\text{C}$
1			
2			
3			

Δ – temperature measurement error in the heater unit, °C;

T_{cert.} – certified reference standard melting point specified in the enclosed certificate, °C.

Conclusion: _____

IV. Self-test:

- In accordance with JP melting point is the temperature at which ...
 - liquid and solid phase of a substance are in an equilibrium.
 - a solid sample is completely melted, ie transferred to the liquid phase.
 - there is a sharp jump in enthalpy ΔH in $\Delta H - T$ curve.
- The melting point determination is used in drugs quality control for ...
 - API identification tests.
 - evaluation of purity of API (impurities limit tests).
 - evidence of chiral purity of drug substance.
- In accordance with JP there are methods to determine the melting point for ...
 - substances those purity is comparably high, which can be pulverized and soluble in water.
 - substances are insoluble in water and hardly dispersible – fats, fatty acids, waxes, waxes.
 - paraffin and petroleum.
 - the nature of the substance does not matter.
- JP Method 1 for melting point determination ...
 - is applied for those substances of which the purity is comparably high and which can be pulverized, soluble in water.
 - involves the use of thermometer with an immersion line.
 - involves the use calibration high-purity substances: acetanilide, acetofenitidine, caffeine, sulfanilamide, sulfapyridine, vaniline.
 - involves the use of thermometers with different temperature ranges: 40–100, 90–150, 140–200, 190–250, 240–320 °C.
 - involves different accuracy in different temperature ranges – from up to $\pm 0,5 \pm 1,0$ °C.
- JP Method 2 for melting point determination ...
 - is applied to substances, that are insoluble in water which can not be pulverized such as fats, fatty acids, paraffin or waxes.
 - involves the use of open capillary.
 - pre-melting the substance at the lowest possible temperature, substance should be placed in ice for 1 hour or stored at temperatures below 10 °C during the day.
 - pre-heating the substance to a temperature of about 5 °C below the expected melting temperature, followed by heating at a rate of not more than 1 °C/min.
- JP Method 3 for melting point determination ...
 - is applied to substances such paraffins or waxes.
 - use of vessel with liquid, in which sample with a thermometer are placed, and a heating vessel.
 - pre-heating to 30 °C at a rate of 2 °C per minute, and further heating at a rate of not more than 1 °C per minute.
 - implies melting point is temperature at which the first drop of the sample leaves the thermometer.

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Laboratory practice session 04

OPTICAL METHODS FOR MEDICINES QUALITY CONTROL. POLARIMETRY

I. Questions and tasks for discussion:

1. Why doesn't a racemic mixture show optical activity?
2. Does the following image show optically active molecules? Why?

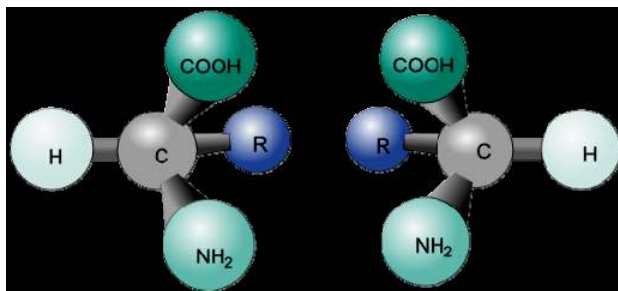


Figure 1

3. What would happen if your feet were not optical isomers of each other?
4. Your friend has tuberculosis. Assuming you like your friend and you want him to live, which isomer of ethambutol do you give him and why?

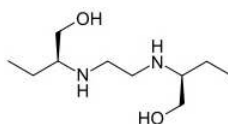


Figure 2

5. Draw each compound's enantiomer:

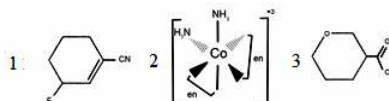


Figure 3

II. Do the tasks:

? Exercise 1

Determine the optical purity of a racemic mixture: $[\alpha]_{\text{pure samples}} = \pm 23,1^\circ$.

Answer

Add texts here. Do not delete this text first.

The specific rotation, $[\alpha]$, of the racemate is expected to be 0, since the effect of one enantiomer cancels the other out, molecule for molecule.

Optical purity, $\omega\% = 100\% [\alpha]_{\text{mixture}} / [\alpha]_{\text{pure sample}} = 100\% (0) / +23,1^\circ = 0\%$.

? Exercise 2

Determine the enantiomeric excess of the racemic mixture.

Answer

Add texts here. Do not delete this text first.

You would expect $[R] = [S] = 50\%$.

$$\omega\% = 100\% ([R] - [S]) / ([R] + [S]) = 100\% (50 - 50) / (50 + 50) = 0\%$$

? Exercise 3

Which isomer is dominant and what is the optical purity of a mixture, of (R)- and (S)-2-bromobutane, whose specific rotation was found to be $-9,2^\circ$? ($[\alpha]_{\text{pure R-sample}} = -23,1^\circ$).

Answer

Add texts here. Do not delete this text first.

The negative sign tells indicates that the R enantiomer is the dominant one.

$$\text{Optical purity, } \omega\% = 100 [\alpha]_{\text{mixture}} / [\alpha]_{\text{pure sample}} = 100 (-9,2) / -23,1 = +40\%.$$

This indicates a 40% excess of R over S.

? Exercise 4

Given that (S)-2-bromobutane has a specific rotation of $+23,1^\circ$ and (R)-2-bromobutane has a specific rotation of $-23,1^\circ$, what is the optical purity and % composition of a mixture whose specific rotation was found to be $+18,4^\circ$?

URL: <http://www.mhhe.com/physsci/chemistry/carey/student/olc/graphics/carey04oc/ch07/others/ch07ans.html>

Answer

Add texts here. Do not delete this text first.

The positive sign indicates that the (S)-isomer is in excess.

$$\text{Optical purity, } \% = 100 [\alpha]_{\text{mixture}} / [\alpha]_{\text{pure sample}} = 100 (+18,4) / +23,1 = 80\%.$$

This indicates a 80% excess of S over R. The 20% leftover, which is optically inactive, must be equal amounts of both (R)- and (S)-bromobutane. The excess 90% is all S so there is a total of 10% (R) and 90% (S).

III. Lab practice.

In this experiment a sugar solution of known concentration (C), but unknown identity will be prepared (note the units of concentration used). The observed rotation (α_{obs}) will be obtained by using a polarimeter. This data will be used to calculate the specific rotation $[\alpha]$ and the identity of the sugar will be hypothesized:

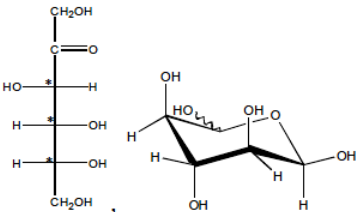
[equation]

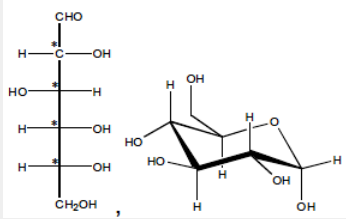
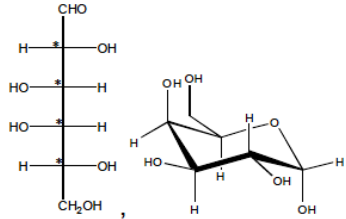
where $[\alpha]$ = specific rotation of the compound;

α_{obs} = observed rotation of light in degrees;

C = concentration in grams per milliliter (g/ml);

l = cell length in decimeters (dm).

Name	Structure (Fisher, Haworth)	Specific Rotation $[\alpha]$
D-Fructose		-86

Name	Structure (Fisher, Haworth)	Specific Rotation [α]
D-Glucose		+98
D-Galactose		+82
Sucrose	Glucose-fructose	+64,5

Sample Preparation, Set Up and Procedure:

1. Weigh out approximately 0,5 g of unknown sample (m). Record all the digits of this mass in your notebook.
2. In 50 ml beaker, dissolve your material in approximately 10 ml of deionized water. Swirl the contents until all of the solid has dissolved.
3. Carefully transfer this solution to a 25 ml volumetric flask. Rinse beaker with approximately 2 ml of deionized water. Transfer this solution to volumetric flask. Repeat.
4. Carefully drop wise add deionized water to volumetric flask until bottom of meniscus is exactly on line. This is your solution.
5. In notebook, calculate the concentration in g/ml.
6. Obtain the α_{obs} by analyzing your solution in the polarimeter using the instructions mounted adjacent to machine and reviewed by your instructor. Repeat the measurement for 3 times.
7. Using the above equation, calculate the specific rotation [α].
8. Select the identity of your material from the list provided.
9. Fill out the relevant spaces in the data table.

Nº	α_{obs}	α_{obs}	l	m	C	[α]	Identification result
1							
2							
3							

Conclusion: _____

IV. Self-test:

1. The Pharmacopoeia prescribes determination of the specific rotation $[\alpha]_{\lambda}^t$ for lots of pharmaceuticals. Which of the following equations is correct?
 - a. [equation]
 - b. [equation]
 - c. [equation]
 - d. [equation]
 - e. [equation]

where $[\alpha]_{\lambda}^t$ = specific rotation in 10 dm/kg;

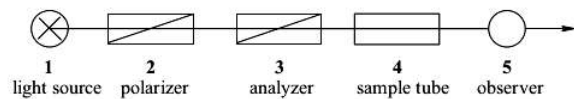
α_{λ}^t = rotation measured at t °C, at wavelength λ ;

l = length of the cell in dm;

ρ_B = mass concentration of substance B in g/0,1 dm;

ρ_t = density of liquid at t °C.

2. Which parts of the polarimeter, if any, should be transposed?



- a. 2 and 4
- b. 3 and 4
- c. 2 and 3
- d. 4 and 5
- e. the sequence of the parts is correct

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Laboratory practice session 05

OPTICAL METHODS FOR MEDICINES QUALITY CONTROL. REFRACTOMETRY

I. Questions and tasks for discussion:

1. Define refraction.
2. Define refractive index.
3. What is the unit of refractive index?
4. List out the factors on which the refractive index of a medium depends.
5. Define angle of incidence.
6. What is the angle of incidence if a ray of light is incident normal to the surface separating the two media?
7. Find the velocity of the light in a medium having refractive index $n = 2$. Speed of light is 300 000 km/sec.
8. Relative refractive indexes of mediums are $n_{1,2} = 2$; $n_{2,3} = 1,5$. Find $n_{1,3}$.
9. If the ray follows following path while passing from one medium to another, find the refractive index of medium.

[image]

II. Lab practice. Determination of magnesium sulfate aqueous solutions concentration by the means of Abbe type refractometer.

Device of Abbe type refractometer is represented on **Fig. 5.1**.

[Fig. 5.1] [A – The appearance of refractometer IRF-454: 1 – ocular, 2 –measuring prism, 3 – light prism, 4 – mirror of illumination scale, 5 – mirror of light illuminating prism, 6 – cylinder removing color, 7 – cylinder displacement of the boundary light-shadow; B – The main working parts of the refractometer: 1 – liquid under investigation, 2 – lighting prism, 3 –measuring prism, 4 – the incident beam, 5 – the field of view]

Starting the work carry out the following steps:

1. Open the window of prism block. Rotating the mirrors illuminate the prism with white light.
2. Customize the refractometer with distilled water. To do this, take up the lighting lens, drop 2–3 drops of distilled water on the measuring prism, close the block.

WARNING! Do not touch the surface of a prism with a pipette.

The whole field of the eyepiece should be illuminated uniformly. Unequal illumination, the dark stains on it point to an insufficient amount of introduced liquid.

Rotate the prism to achieve the appearance of dark field in the eyepiece. The appearance of the dark-field corresponds to the position of the prism in which the light beam undergoes internal reflection from the interface between the prism and the test substance. If the border of dark field is not sharp, then rotate compensator until a sharp boundary of dark field appears. Then move dark field boundary to the intersection of lines. Count the value of the refractive index on the scale.

4. Open the box and wipe the prism surface with a soft cloth or filter paper.
5. Place 2–3 drops of the standard solution between the two halves of the prism and tightly compress them. Combine sight buttons with the boundary line separating the light and dark parts of the visual field. Record the scale division of the refractive index, which coincides with the sighting stroke. Measure the refractive index three times for each solution.

Before each subsequent measurement the prism should be thoroughly cleaned with distilled water and then wiped with filter paper.

After measuring the refractive index of 4–5 solutions of magnesium sulfate place the results in **Table 5.2**.

Plot the dependence of the refractive index n of the solution concentration C (x – values C_i , and Y –values n_i). Don't forget: the value of the refractive index of distilled water = 1,33. Draw a straight line through the data points.

The measurements results of the refractive indices of magnesium sulfate solutions

N_0	n_1	n_2	n_3	n_{cp}
N_0 1. $C = __\%$				
N_0 2. $C = __\%$				

Nº	n ₁	n ₂	n ₃	n _{cp}
Nº 3. C = ____ %				
Nº 4. C = ____ %				
Nº 5. C _x = X %				

[image]

Find the value $K(F)$ – the refractive index increment ($dn/dC = \text{tg}\gamma$, where γ – the angle of the slope to the horizontal axis). Substituting the value of K in the formula $n = n_0 + K \cdot C$ find the concentration of the solution N^o 5:

[equation]

where n – refractive index of solution N^o 5,

n_0 – refractive index of water.

III. Self-test:

- Refractometry in pharmaceutical analysis is used for:
 - identification
 - purity assessment
 - assay
- The basis of refractometry is the law:
 - of light absorption
 - of light refraction
 - of light scattering
 - of light polarization
- The ratio of light speed in vacuum to its speed in the medium is called:
 - the relative refractive index
 - the absolute refractive index
- The refractive index depends on:
 - temperature
 - the wavelength of light
 - the length of the cuvette
 - solution concentration
 - the nature of the solvent
- In the method of refractometry the concentration of sample solution is calculated by the formula:
 - [equation]
 - [equation]
 - [equation]
 - [equation]
- Specify a range of solutions refractive indices, which can be measured in daylight:
 - 0,2–0,8
 - 1,3–1,7
 - 1,8–2,5
- Designation of the refractive index by symbol n_D^{20} indicates that:
 - The measurement was performed at 20 °C.
 - The measurement was performed at the wavelength of the yellow line of the sodium spectrum.
 - The measurement was performed at a wavelength of the red line of the hydrogen spectrum.
 - The measurement was performed at a wavelength range of the blue line of hydrogen.
 - The measurement was performed at wavelength of violet spectral lines hydrogen.

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